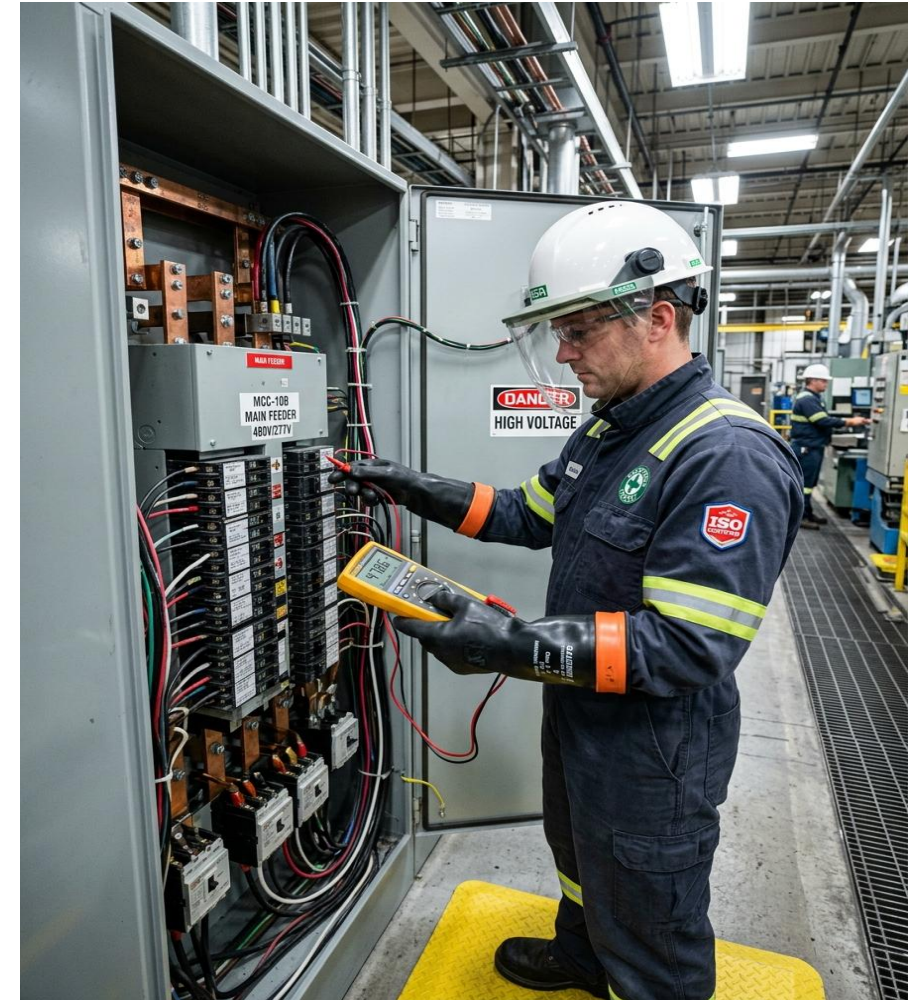


TRAINING

Electrical Safety

Practical basics of safely operating electrical installations, technical safeguards and admission to work

Training course for electrical and electrotechnological personnel and line supervisors



After the course you can

- 01** — Recognise electric current hazards, assess risks and keep safe distances.
- 02** — Correctly use earthing, RCDs, dielectric equipment and portable earths.
- 03** — Flawlessly perform technical and organisational measures to prepare the worksite and admit crews.
- 04** — Act instantly in an electrical injury: free the victim from the current and give first aid.

**The course's main goal
— zero injuries and no
contact with current.**

Four course parts

- 01 — Electrical safety basics and effects of electric current.
- 02 — Technical safeguards and electrical protective gear.
- 03 — Safe work organisation, electrical safety groups and permits.
- 04 — Freeing a victim from current and first aid.

COURSE

4

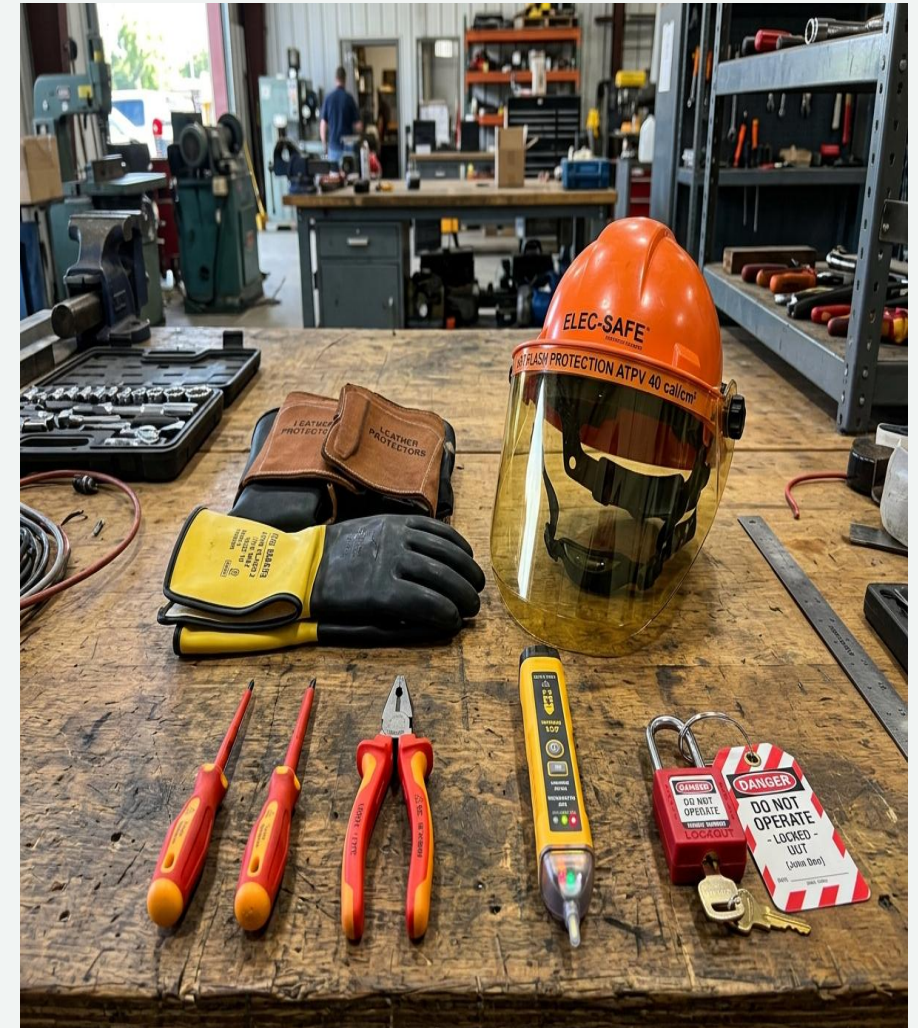
practical modules of
electrical safety



Electrical safety basics

Electric current is invisible, odourless and acts instantly. The only protection is knowing the laws and strictly following the rules.

Reference: PUE RK • Safety Rules for the operation of electrical installations





Electrical safety at work

- **Work environment:** electrical installations up to and above 1000 V, switchboard rooms, cable routes and switchgear.
- **Nature of the hazard:** electric current shows no outward signs of danger (no colour, smell or sound in normal operation).
- **Human factor:** loss of vigilance, getting used to danger and ignoring the rules cause over 80% of severe electrical injuries.
- **Main rule:** every electrical installation is considered live until proven otherwise and until earthing is in place.

Electric current forgives no mistakes: the lack of visible danger dulls your vigilance!

Effects of electric current on the human body

TYPE OF EFFECT	PATHOLOGICAL EFFECT	CONTROL AND PROTECTION
Thermal	Severe tissue burns, vessel heating, boiling blood and lymph	Insulation, protective shields, arc-rated PPE
Electrolytic	Decomposition of body fluids, blood breakdown, chemical changes	De-energising, dielectric gloves and insulated tools
Mechanical (dynamic)	Torn muscles, tendons and blood vessels, fractures, dislocations	Automatic disconnection (RCD, fast-acting relays)
Biological	Muscle spasms, cardiac fibrillation, respiratory arrest	Equipotential bonding, TN neutral connection, earthing

Severity depends on the size of the current, the exposure time and its path through the body.



4 thresholds of alternating current (50 Hz)

1

0.6 — 1.5 mA

Perceptible current: slight tingling, faint itch in the fingers.

2

10 — 15 mA

Non-let-go current: irresistible muscle spasms, the hand cannot unclench.

3

50 — 100 mA

Fibrillation current: chaotic heart contractions, circulatory arrest.

4

> 5 A

Lethal current: instant heart paralysis, the most severe thermal burns.

A current of just 0.1 A (100 mA) for 1–2 seconds is fatal!

Local electrical injuries and effects

- **Electrical burns:** contact current burns (current passing through tissue) and arc burns (arc temperature reaches 4000–6000 °C).
- **Electrical marks:** firm grey or yellowish spots of dead skin with sharp edges at the current entry and exit points.
- **Skin metallisation:** tiny splashes of molten metal driven into the epidermis by an electric arc.
- **Electro-ophthalmia:** inflammation of the cornea and conjunctiva from intense arc UV radiation (goggles or a face shield required).

Electric arc temperature reaches 6000 °C — working without a face shield and arc suit is deadly!



Current paths through the body (current loops)

- **Danger of a current loop:** set by the share of total current passing through vital organs — the heart, lungs and brain.
- **“Hand — hand” loop:** ~3.3% of the current passes through the heart. The most common workplace case when touching two phases.
- **“Hand — feet” loop:** more current reaches the heart than in the “hand — hand” loop: ~3.7% from the left hand and ~6.7% from the right.
- **“Foot — foot” loop:** step voltage current; ~0.4% passes through the heart, but falling to the ground turns it into “hands — feet”.

A fall in the current zone instantly turns the loop from “foot-foot” to the deadly “hands-feet”!

CRITICAL BAN: Dangerous approach distances



- Do not approach unguarded live parts closer than the set minimum distances (up to 1 kV — 0.6 m; 1–35 kV — 0.6 m; 110 kV — 1.0 m; 220 kV — 2.0 m).
- Do not work alone in electrical installations where the work permit requires a second person to be present.
- Do not touch live insulators and structures without basic electrical protective equipment.
- Do not continue work if there are signs of arcing, an unusual transformer hum or a smell of burning.

A person or tool coming closer than permitted causes flashover across the air gap!

Premises classified by electrical hazard

- **Premises without increased danger:** dry, normal temperature, insulating wooden floors, no conductive dust.
- **Premises with increased danger:** damp (humidity >75%), conductive dust, conductive floors (metal, concrete), temperature >35 °C.
- **Especially dangerous premises:** extreme damp (humidity near 100%), chemically aggressive environment or two signs of increased danger at once.
- **Outdoor switchgear areas:** always equated to the category of especially dangerous premises!

Any work outdoors or in the open air is automatically treated as work in especially dangerous conditions!

Touch voltage and step voltage

- **Step voltage:** the potential difference between two points on the ground one human step apart (0.8 m).
- **Current spreading zone:** occurs when an OHL wire falls to the ground or on an earth fault. Danger zone radius — 8 metres (indoors 4–5 m).
- **Potential distribution:** falls hyperbolically from the fault point outwards; the largest difference is near the fallen wire.
- **No hopping on one foot:** losing balance and falling onto your hands is deadly, as it creates a “hands — feet” current loop.

**The danger radius
around a fallen wire is 8
metres!
Never approach a wire
on the ground!**



4 rules for leaving a step voltage zone

1

Feet together

Immediately bring your feet together so there is no potential difference.

2

“Shuffle”

Move slowly, with the heel of one foot against the toe of the other, never lifting.

3

Step ≤ 10 cm

Step length must not exceed the foot's width; no wide steps!

4

Get 8 m away

Keep moving until fully out of the 8-metre danger radius around the fallen wire.

Never run or jump — losing your balance will be fatal!



Section I summary: 5 electrical safety barriers

- **1. Hazard awareness:** remember that current is invisible and never be careless in any operation on electrical installations.
- **2. Distance barrier:** strictly keep the minimum permissible distances to live parts.
- **3. Environmental control:** account for humidity, conductive floors and open structures, which cut body resistance tenfold.
- **4. Behavioural rules:** no working alone where the rules require a crew or supervision.
- **5. Incident readiness:** solid knowledge of step voltage and the first aid algorithm for a victim.

Electrical safety is not a set of tips but the law of survival on an industrial site!

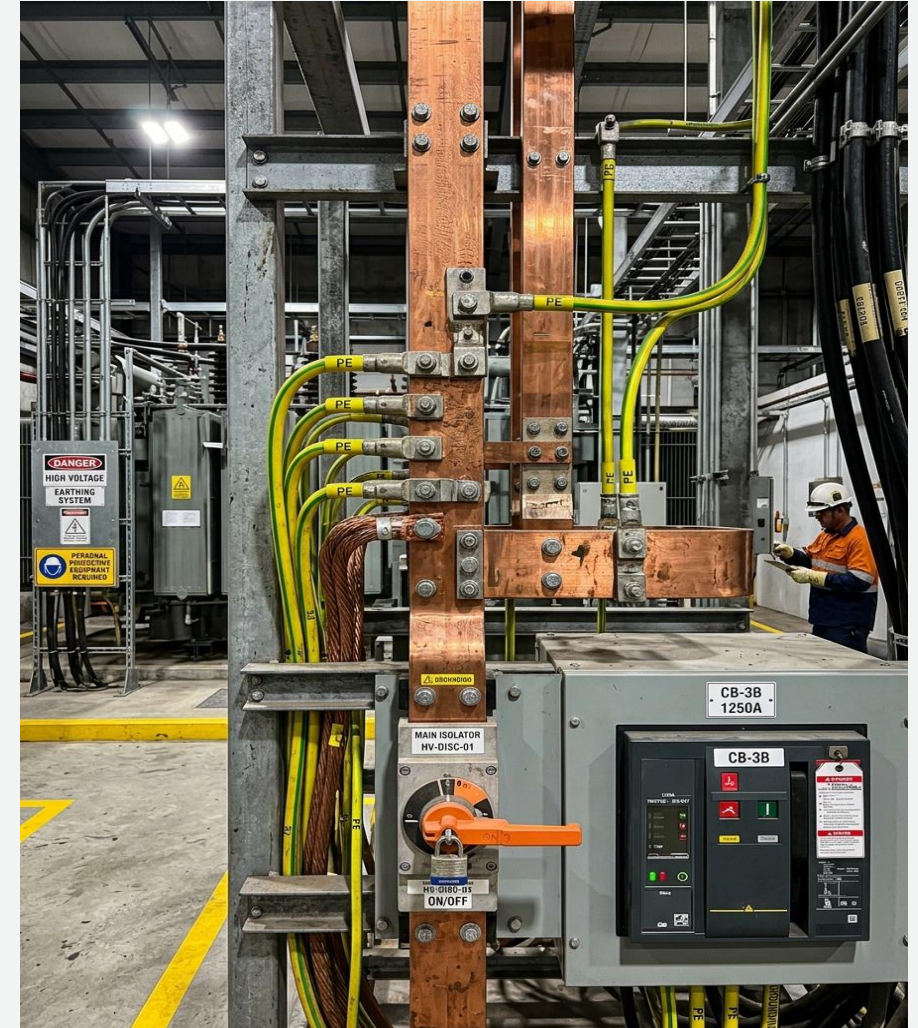


Technical safeguards

Reliable earthing, insulation, extra-low voltage and electrical protective equipment form an impassable technical barrier.

Reference: PUE RK Chapter 1.7 • GOST 12.1.019-2017

Electrical safety (ES) • SafetyHUB



Protective earthing: how it works

- **Definition:** a deliberate electrical connection of the non-current-carrying metal parts of an installation to an earthing device.
- **Purpose:** to reduce touch voltage to a safe level when phase conductor insulation breaks down to a metal enclosure.
- **Physical principle:** the earth electrode resistance ($R_e \leq 4 \text{ Ohm}$) and the human body ($R_b \sim 1000 \text{ Ohm}$) in parallel.
- **Result:** most of the fault current flows to earth through the low-resistance earthing loop, protecting the person.

Current takes the path of least resistance: good earthing takes the hit!

Earthing systems of power networks

- **TN-S system:** the neutral (N) and protective (PE) conductors are separate throughout the network — the safest system.
- **TN-C-S system:** N and PE functions are combined in one PEN conductor in the upstream part and split at the building entry.
- **TN-C system:** an outdated system with a combined PEN throughout; prohibited for new and reconstructed installations!
- **TT and IT systems:** TT — independent earthing of the installation; IT — isolated neutral with insulation monitoring (mines, mobile units).

Separating PE and N conductors is the key to correct RCD operation and electrical safety!

Protective neutral (TN): purpose and tripping

- **TN connection:** a deliberate electrical connection of exposed conductive parts to the solidly earthed neutral of the supply via the PE conductor.
- **Protection principle:** turns a phase-to-enclosure fault into a short circuit (SC) between the phase and the protective conductor.
- **Protection tripping:** the short-circuit overcurrent instantly trips the circuit breaker or blows the fuse link.
- **Standard time:** the faulty section of a 220/380 V network must be disconnected within 0.4 seconds (per the PUE).

TN connection turns a fault into a short circuit and cuts the faulty section in a split second!

Residual current devices (RCD)

- **Differential transformer:** continuously compares the vector sum of currents entering the load via phase and returning via neutral.
- **Leakage current:** when a person touches a live part, a residual current (earth leakage) arises and induces an EMF in the secondary winding.
- **Tripping threshold:** to protect people against direct contact, RCDs with a rated residual current $I_{\Delta n} \leq 30 \text{ mA}$ are used.
- **Response speed:** a quality RCD disconnects the faulty circuit in 0.02–0.04 seconds, before heart fibrillation sets in.

A 30 mA RCD is the only protection that can save a life in direct contact with a bare wire!



Extra-low voltage (ELV / SELV)

- **Definition:** rated voltage not exceeding 50 V AC or 120 V DC between conductors or to earth.
- **Application:** supply of portable hand-held power tools and portable lamps in conditions of increased danger.
- **Voltage levels:** 42 V (or 36 V) — in premises with increased danger; 12 V — in especially dangerous conditions (inside boilers, tanks, wells).
- **Power source:** only via isolating step-down transformers with separate windings (autotransformers are prohibited!).

Inside metal tanks and wells, only lamps rated at no more than 12 volts may be used!

Electrical protective equipment types

- **Basic up to 1000 V:** dielectric gloves, tools with insulated handles, voltage indicators, clamp meters.
- **Additional up to 1000 V:** dielectric overshoes, dielectric mats, insulating stands and covers, protective barriers.
- **Basic above 1000 V:** insulating rods of all types, insulating tongs, voltage indicators above 1000 V.
- **Additional above 1000 V:** dielectric boots, dielectric gloves (above 1 kV they are only additional!), mats.



Basic PPE withstands the operating voltage; additional PPE is used only together with basic PPE!



Electrical protective equipment: 4 checks

1

Test stamp

Check the next test date on the stamp.
Expired PPE is prohibited!

2

Visual check

Make sure there are no cracks, punctures, tears, chips or mechanical defects.

3

Roll test

Roll the glove towards the fingers to check for punctures under air pressure.

4

Clean and dry

Before work, wipe off dust, dirt, oil and moisture with a clean, dry cloth.

Using electrical protective equipment with an expired test stamp is strictly prohibited!



4 steps to verify absence of voltage

1

Self-check

Check that the indicator works on parts known to be live or with a test device.

2

Phase by phase

Verify there is no voltage between each phase and earth, then between all phases.

3

Induced voltage

Make sure there is no dangerous induced voltage from adjacent energised lines and phases.

4

Re-test

Re-check the indicator on a known live source to rule out a sudden failure.

Always wear dielectric gloves when verifying absence of voltage!



Portable protective earthing

- **Purpose:** protect workers from voltage applied in error by switching devices and from induced voltage.
- **Fitting order:** FIRST connect the earth clamp to the earthing device (loop), and only THEN to the live parts.
- **Removal order:** strictly in reverse: FIRST take the clamps off the live parts, and only THEN detach the clamp from the earth electrode.
- **Earthing conductor cross-section:** flexible copper wires of at least 16 mm² (up to 1000 V) and at least 25 mm² (above 1000 V) for thermal withstand.

Fitting earths: first to Earth, then to Phase!
Removing: first from Phase, then Earth!
Wrong order is deadly!

Electrical safety signs and posters

- **Prohibition:** “DO NOT SWITCH ON! PEOPLE AT WORK”, “DO NOT OPEN! PEOPLE AT WORK” — hung on the drives of switching devices.
- **Warning:** “STOP! HIGH VOLTAGE”, “DO NOT CLIMB! DANGER OF DEATH”, “CAUTION! ELECTRICAL HAZARD” — mark off the danger zone.
- **Mandatory:** “WORK HERE”, “CLIMB HERE” — clearly mark the prepared and verified safe workplace.
- **Indicative:** “EARTHED” — hung on device keys and control handles and on earthed sections of the installation.

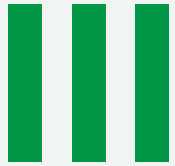
The “DO NOT SWITCH ON” sign has the legal force of an order!
Only the person who hung it may remove it!

CRITICAL BAN: Breaching technical measures



- Do not start work in an electrical installation without completing all the mandatory technical measures to prepare the site.
- Do not switch on disconnected devices or remove prohibition signs and mechanical locks without authorisation.
- Do not use random wires of non-standard cross-section or twisted wires as portable protective earths.
- Do not touch live parts without verifying absence of voltage with a working indicator while wearing dielectric gloves.

De-energising without earthing does not guarantee safety — induced voltage kills just like operating voltage!



Work organisation and permits

Clearly divided responsibility, strict admission rules and the work permit rule out the human factor and errors.

Reference: Safety Rules of the RK • Safety Rules • Admission to work

Electrical safety (ES) • SafetyHUB



Two ways to authorise electrical work



WORK PERMIT

- A written order on a prescribed form.
- Sets the exact place and the start and end times of work.
- Records the crew, responsible persons and safety measures.
- Required for de-energised work and work near live parts.
- Maximum validity — no more than 15 calendar days.



WORK ORDER

- A verbal task for a one-off simple job.
- Valid for one work shift of the crew.
- Must be recorded in the site operational log.
- Done by a small team (1–2 people) with a clear scope.
- Used for unplanned urgent minor work up to 1 kV.

Routine operational work follows a pre-approved list, without a work permit or work order.



Electrical safety qualification groups

- **Group I:** non-electrical staff (office, warehouses, PC operators); briefing once a year, recorded in the log.
- **Group II:** electrotechnological staff (welders, crane operators, machine operators); 72+ hours of training, knows installations up to 1 kV.
- **Group III:** electrical staff; 1–3 months' experience in Group II, may inspect alone and work up to 1 kV.
- **Group IV:** may work in installations above 1 kV and act as work supervisor, permit issuer or person in charge of electrical facilities up to 1 kV.
- **Group V:** highest level (engineers, energy managers, persons in charge of installations above 1 kV); knows all circuits and rules.

A safety group is a personal measure of responsibility and the right to independent access to hazardous work!

Persons responsible for safe work

- **Permit issuer (work order giver):** decides if work is needed, its safe scope and the qualifications of the crew's responsible persons.
- **Responsible work manager (RWM):** ensures the planned safety measures are complete and sufficient and coordinates the crews.
- **Permitting officer:** responsible for correct workplace preparation, checking absence of voltage, fitting earths and signs.
- **Work supervisor (WS):** responsible for a clear on-site crew briefing, correct use of PPE and continuous supervision.
- **Observer and crew members:** the observer supervises non-electrical staff; crew members strictly follow the WS's instructions.

Each responsible person answers for their part of preparation — combining roles is strictly regulated by the SR!



6 steps to prepare the workplace

- **1. Disconnection:** make the required disconnections with a visible break in the circuit.
- **2. Locks and signs:** lock the drives with LOTO locks and hang prohibition signs.
- **3. Voltage check:** verify absence of voltage on all phases with a working indicator.
- **4. Earthing:** close the fixed earthing switches or apply portable earths.
- **5. Indicative signs:** hang “EARTHED” signs on device keys and handles.
- **6. Fencing:** fence off the workplace with portable screens and hang safety signs.

Strict order of the 6 preparation steps:
disconnect → lock → sign → test → earth → fence!

LOTO protective lockout in power networks

- **LOTO concept (Lockout / Tagout):** mechanical isolation of hazardous energy sources with personal padlocks and warning tags.
- **Personal lock:** each crew member puts their own lock on the breaker or switch lockout device and keeps the only key.
- **Group lockout:** with several crews, a group lock box is used: equipment keys go inside, and every team puts its lock on the box.
- **Error-proofing:** no breaker can be switched on remotely or manually until the last worker removes their personal lock.

**One worker — one lock
— one key!**
**No one can energise the
line while you are
working on it!**



4 steps to admit a crew under a work permit

1

Staff check

Check the crew by names, ID certificates and electrical safety groups.

2

Show on site

Prove absence of voltage, show the earths, boundaries and adjacent equipment.

3

Task briefing

Explain the work method, danger zones, escape routes and safe work practices.

4

Sign the permit

Permitting officer, work supervisor and crew sign the permit in person.

No work may start without the permitting officer's and work supervisor's signatures on the permit!

Continuous supervision during work

- **Duty of the work supervisor (observer):** continuously supervise all crew members throughout the task.
- **No leaving:** the work supervisor must not combine supervision with any other work or leave the workplace.
- **Supervisor's temporary absence:** if the work supervisor has to leave, the entire crew must be withdrawn from the installation.
- **Crew members:** may not stay in the switchgear or at height without the work supervisor; switchgear doors are locked during breaks.

**The work supervisor is the crew's “safety eyes”:
doing their own work
means losing control over
people!**

Breaks and completion of work

- **Lunch break:** the crew leaves the work zone; the permit stays with the work supervisor; signs, earths and barriers stay in place.
- **Re-admission after a break:** the work supervisor personally checks that the workplace is unchanged and admits the crew without the permitting officer.
- **Daily end of work:** the workplace is tidied up and the permit is handed to the duty operating staff, marked as end of shift.
- **Final completion:** temporary fixtures are dismantled, crew earths removed, and the permit is closed with WS and permitting officer signatures.

After the shift, switching on the installation before the work permit is formally closed is strictly prohibited!



4 work supervisor actions during a break

1

Withdraw crew

Stop the tasks, collect all tools and lead people out of the electrical installation.

2

Lock access

Lock the switchroom doors, barriers and gates of the work zone.

3

Keep the setup

Leave all signs, LOTO locks and portable protective earths in place.

4

Check before re-entry

Before resuming, personally confirm the setup is safe and the earths are intact.

Crew members must not enter the work zone before the work supervisor returns!

Admission of seconded staff and contractors

- **Secondment letter:** the employer sends an official letter listing the workers, their safety groups and the rights of responsible persons.
- **Knowledge check:** seconded staff receive induction and initial briefings on the client's electrical installation layout.
- **Admission certificate:** issued when a separate fenced work zone is allocated to the contractor on an operating site.
- **Assigning the permitting officer:** initial admission of seconded crews is always done only by the client's own operating staff.

A contractor on site calls for extra attention: initial admission is done only by the owner company's own staff!

Section III summary: Organisational control chain

- **Unbroken chain:** safety is guaranteed only when every link works — permit, briefing, LOTO and supervision.
- **Personal responsibility:** each worker signs the work permit for their own and their colleagues' safety.
- **Documenting changes:** any change in the crew or work conditions requires the admission to be reissued at once.
- **Life comes first:** organisational rules are based on the analysis of hundreds of real incidents and are mandatory.

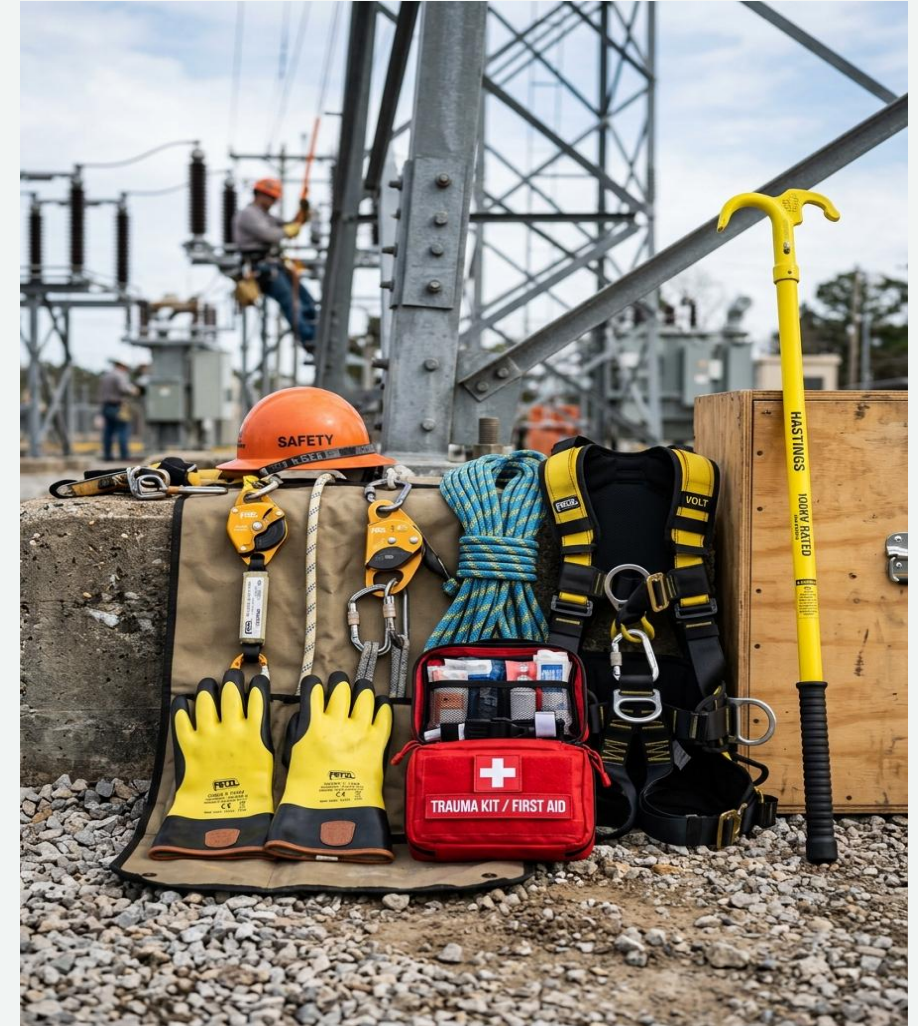
A work permit is not a formality but a legal document that protects your life and freedom!

IV

Help with electrical injuries

Fast, correct release from the current and immediate resuscitation save the victim's life.

Reference: Cross-industry first aid instruction • Electrical injuries



Releasing a victim from current at height

- **Step 1. De-energise the installation:** switch off the nearest isolator, breaker or switch with the emergency stop.
- **Step 2. Prevent a fall:** remember that once the current is off, the victim will go limp and fall unless secured by fall protection!
- **Step 3. Insulate the rescuer:** if power cannot be cut quickly, use dielectric gloves, an insulating rod or an axe with a dry handle.
- **Step 4. Separate the victim:** pull the victim by dry clothing only, with one hand, without touching bare skin or metal.

When cutting the current to a victim at height, secure them from falling: the fall after a shock is deadlier than the current!



First aid: cardiopulmonary resuscitation (CPR)

- **Assessment:** check consciousness (call out, shake the shoulders) and breathing (the “listen — look — feel” rule for 10 seconds).
- **Chest compressions:** 30 presses on the centre of the breastbone, 5–6 cm deep, at 100–120 per minute with the heel of the hand.
- **Rescue breaths:** 2 mouth-to-mouth breaths through a barrier face shield after tilting the head back and pinching the nose.
- **Using an AED (defibrillator):** apply the automated defibrillator pads and follow the device's voice prompts.
- **Fractures and bleeding:** apply a pressure bandage to bleeding, splint fractures, do not reset bone fragments!

**Start CPR immediately!
Every minute of delay cuts
the brain's chance of
survival by 10%!
Ratio: 30 to 2!**



CRITICAL STOP: 5 grounds for an absolute stop



- 1. Sudden appearance of voltage on disconnected live parts or sparking at the worksite.
- 2. Detection of a defect or an expired test stamp on electrical protective equipment.
- 3. Sudden illness of any crew member (dizziness, nausea, weakness, chest pain).
- 4. Forced departure of the work supervisor or the person providing continuous supervision.
- 5. An emergency, or tools or structures falling near people.

The right to stop work (“STOP CARD”) is every worker's sacred right, with no risk of punishment!



Final admission checklist: 6 safety conditions

- ✓ **Work permit issued** — Permit signed by all parties, still valid, task briefing given.
- ✓ **De-energised and earthed** — Switching devices off, absence of voltage verified, earths applied.
- ✓ **LOTO lockout in place** — Personal mechanical locks on the drives, keys held by crew members.
- ✓ **Electrical PPE checked** — Dielectric gloves, tools and voltage detectors tested, stamps valid.
- ✓ **Signs posted** — “DO NOT SWITCH ON! PEOPLE AT WORK” on the drives, “EARTHED” on the handles, workplace fenced off.
- ✓ **First aid ready** — Crew knows how to free a victim from current and do CPR, first aid kit on site, communications working.